

Camilo Uribe Guerra

Data Engineer · AI Practitioner · Energy & Sustainability

Master of Data Science and Innovation

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in LinkedIn Profile

GitHub Profile

SUMMARY

Data engineer and data scientist with 3+ years of production experience delivering ETL pipelines, ML models, and client-facing analytics solutions across energy and SaaS environments; petroleum engineer by background with depth in ESG and GHG emissions analytics. Strong technical foundation in Python, SQL, PySpark, dbt, and Snowflake — with a track record of direct client delivery and recent hands-on application of generative AI, RAG pipelines, and agentic systems. Currently interning as a Data Scientist at Synogize while completing a Master of Data Science and Innovation at UTS (GPA 6.7/7.0) — committed to contributing to Australia's digital and energy transition.

EDUCATION

- **Master of Data Science and Innovation — GPA 6.7/7.0**
University of Technology Sydney, Sydney, Australia Expected Graduation: July 2026
- **Specialization Degree in Data Analytics — GPA 4.8/5.0**
Universidad Nacional de Colombia, Medellín, Colombia Jan 2022 – Apr 2023
- **Bachelor of Science in Petroleum Engineering — GPA 4.5/5.0**
Universidad Nacional de Colombia, Medellín, Colombia Jan 2016 – Apr 2021

EXPERIENCE

- **Data Scientist & AI Intern**
Synogize, Sydney, NSW, Australia Feb 2026 – Present
 - Contributing to Synogize's Pentana dealership data enablement project by helping turn raw ERA Power operational data into a reusable analytics foundation for reporting and downstream business use cases.
 - Executed cleaning, restructuring, validation, and data preparation for a large dealership sales dataset supporting Snowflake-based analysis across sales performance, inventory, and profitability.
- **Lead AI Intern**
Triber Tech, Sydney, NSW, Australia Nov 2024 – Jan 2025
 - Supported rollout of AI workflows across product, design, and marketing teams, contributing to an estimated **30%** improvement in operational efficiency.
 - Designed and deployed a retrieval-based resume ranking and supplier evaluation workflow.
 - Supported internal AI adoption through documentation, training, and change management workshops.
- **Data Engineer — Customer Success Engineer**
Tachyus, Houston, TX, USA Mar 2023 – Jan 2025
 - Delivered ETL and analytics workflows for nearly ten clients, integrating data from CSV, Excel, PDF via OCR, SQL, and NoSQL sources to support operational and analytical use cases.
 - Built Python- and SQL-based workflows for data processing, forecasting, GHG calculations, and production optimisation, with a strong focus on reliable outputs and practical business value.
 - Worked directly with customer stakeholders on SaaS implementations, data quality automation, technical demonstrations, and business-facing presentations in a fast-moving startup environment.
- **Data Scientist**
Skanhawk, Bogotá D.C., Colombia Sept 2022 – Feb 2023
 - Conducted real-time and post-mortem analysis for operational performance, achieving a **15–20%** reduction in operational times.
 - Built and tuned analytics models, including an XGBoost model with **85%** accuracy for predicting operation outcomes.
 - Designed and delivered interactive SaaS dashboards for clients using business-focused analytics.
- **Engineering Operations Experience**
Halliburton · Petroland, Colombia 2021 – 2022
 - Early engineering and reporting experience supporting operational workflows and production reporting, providing additional context for data and analytics delivery.
- **Research Assistant**
Universidad Nacional de Colombia, Medellín, Colombia Aug 2018 – Jul 2020
 - Developed reservoir simulation software in C++ and Fortran for CEOR processes — early experience in computational problem-solving and engineering research.
 - Researched nanotechnology applications to optimise surfactants and polymers in enhanced oil recovery.

PROJECTS

Airflow + dbt Medallion Pipeline for Airbnb and Census Analytics: Built an end-to-end Airflow and dbt pipeline using bronze, silver, and gold layers, SCD Type 2 snapshots, star-schema models, and business marts for revenue and demographic analysis.

RAG Resume Ranking and Supplier Evaluation System: Built an AWS Bedrock-based retrieval workflow using MongoDB as the vector store to provide LLM context, supporting resume ranking and supplier evaluation use cases.

Reinforcement Learning GHG Consultant: Led project delivery and business analysis for an RL-based assistant for Australian greenhouse gas compliance, improving answer quality, groundedness, and user satisfaction.

LLM Prompt Injection Security Evaluation Framework: Developing a modular capstone framework to test prompt injection attacks, benchmark attack success, and evaluate mitigation strategies across LLM and lightweight agent workflows.

TECHNICAL SKILLS

Programming: Python (Advanced), SQL (Advanced), R (Intermediate), C++ (Intermediate)

Data Engineering: ETL/ELT pipelines, data modelling, data quality automation, Pandas, PySpark (Databricks), PyArrow, Airflow, dbt, Snowflake, PostgreSQL, MongoDB

Analytics & AI: Machine learning, statistics, forecasting, generative AI, RAG, prompt engineering, LangChain, LlamaIndex, HuggingFace

App & API Development: FastAPI, Streamlit, Dash

Visualization & BI: Power BI, Tableau, Plotly, Matplotlib

Cloud & Deployment: AWS, Azure, GCP, AWS Bedrock, Render

Developer Tools: Git, Docker

Professional Strengths: Stakeholder communication, client-facing delivery, cross-functional collaboration, problem solving, adaptability

CERTIFICATIONS AND COURSES

Agentic AI and Autonomous Agents — IBM (Coursera)

Certified Data Scientist — CorrelationOne DS4A Colombia

Excel Skills for Business Essentials — Macquarie University

Python Data Structures and Accessing Web Data — University of Michigan

PAPERS

Automation of Greenhouse Gas Emissions Estimates Associated with Natural Gas Production in the Appalachian Basin. (OnePetro)

LANGUAGES

Spanish: Native **English:** Advanced

REFERENCES

Available upon request.